

# Automated EEG Signal-Based Sleep State Classification and Analysis

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**Abstract**—This project presents an automated approach for sleep stage classification using electroencephalogram (EEG) signals from the PhysioNet Sleep-EDF Expanded dataset. Manually scoring these signals is time-consuming and subject to variability, motivating the use of machine learning for automation. In this work, classical models (Support Vector Machine and Random Forest) and a 1D Convolutional Neural Network (1D-CNN) are used to classify 30-second EEG epochs into five sleep stages: Wake, N1, N2, N3, and REM. EEG signals were preprocessed and segmented, with handcrafted features extracted for classical models, while the 1D-CNN operated directly on raw EEG signals. Performance was evaluated using accuracy, balanced accuracy, and macro F1-score. Results show that the 1D-CNN outperforms classical methods, particularly in minority classes such as REM, while N1 remains the most challenging stage due to its transitional nature. These findings demonstrate the effectiveness of deep learning for EEG-based sleep staging and highlight areas for improvement, including handling class imbalances and incorporating temporal modeling. Python code, analyses, and results are available online on Github (see References).

**Index Terms**—EEG, Electroencephalogram, Biomedical Signal Processing, Sleep Stage Classification, Machine Learning, Deep Learning, CNN, SVM, Random Forest

## I. INTRODUCTION

Sleep plays a critical role in cognitive function, memory consolidation, and overall health. Sleep studies commonly utilize electroencephalogram (EEG) signals to monitor brain activity and classify sleep stages, which are typically segmented into 30-second epochs and labeled according to clinical standards. These stages include: Wake, N1, N2, N3, and REM, and are characterized by distinct frequency patterns in EEG signals.

Traditionally, sleep stage classification is performed manually by trained experts who visually inspect EEG recordings. This process is time-consuming, labor-intensive, and subject to inter-scorer variability. As a result, there is growing interest in developing automated methods for sleep staging using machine learning techniques.

Prior work in this area has explored both classical machine learning approaches and deep learning models for EEG-based classification. Classical methods typically rely on handcrafted feature extraction, such as frequency band power and statistical descriptors, while deep learning models, particularly convolutional neural networks, can learn features directly from raw EEG signals.

In this work, an automated sleep stage classification system is developed using both classical machine learning models (Support Vector Machine and Random Forest) and a 1D Convolutional Neural Network (1D-CNN). The models are trained and evaluated on the PhysioNet Sleep-EDF Expanded dataset using a Leave-One-Subject-Out (LOSO) cross-validation methodology. The primary objective is to compare the performance of classical and deep learning approaches, with a focus on handling class imbalances and improving classification of more challenging stages such as N1 and REM.

## II. APPROACH

### A. Dataset

The dataset used in this study is the Sleep-EDF Expanded dataset obtained from PhysioNet. It contains overnight EEG recordings from 40 subjects. Each recording includes a single EEG channel (Fpz–Cz) sampled at 100 Hz. The EEG signals are segmented into 30-second epochs, consistent with clinical sleep staging standards, resulting in approximately 3000 samples per epoch. Each epoch is labeled as one of five sleep stages: Wake, N1, N2, N3, and REM.

### B. Preprocessing

To improve signal quality, a band-pass filter between 0.5 Hz and 40 Hz was applied to remove high-frequency noise and movement. Epochs corresponding to movement artifacts or unknown labels were removed from the dataset as well.

For classical machine learning models, feature extraction was performed on each epoch. A total of 11 features were computed, including:

- Frequency band power (delta, theta, alpha, beta)
- Statistical moments (mean, variance, skewness, kurtosis)
- Hjorth parameters (activity, mobility, complexity)

These features are commonly used in EEG signal analysis and capture both frequency-domain and time-domain characteristics of the signal.

### C. Models

Three models were developed and evaluated:

- 1) **Support Vector Machine (SVM):** An SVM with a radial basis function (RBF) kernel was used as a baseline classifier.

- 2) **Random Forest (RF):** An ensemble of decision trees was used to improve classification performance through bagging and feature randomness.
- 3) **1D Convolutional Neural Network (1D-CNN):** A deep learning model was designed to operate directly on raw EEG signals. The architecture consists of three Conv1D layers with increasing filters (32, 64, 128), each followed by batch normalization and max pooling. A global average pooling layer is used to reduce dimensionality, followed by a dense layer with dropout for regularization. The final layer uses a softmax activation function to classify the input into one of five sleep stages.

#### D. Training and Evaluation

Model performance was evaluated using Leave-One-Subject-Out (LOSO) cross-validation for classical models to assess generalization across subjects. To address class imbalance, SMOTE (Synthetic Minority Over-sampling Technique) was applied during training for classical models, while class weighting/balancing was used for the 1D-CNN.

Performance was measured using the following metrics:

- Accuracy
- Balanced Accuracy
- Macro F1-Score
- Per-Class F1-Score

These metrics provide a comprehensive evaluation, particularly in the presence of strong class imbalances.

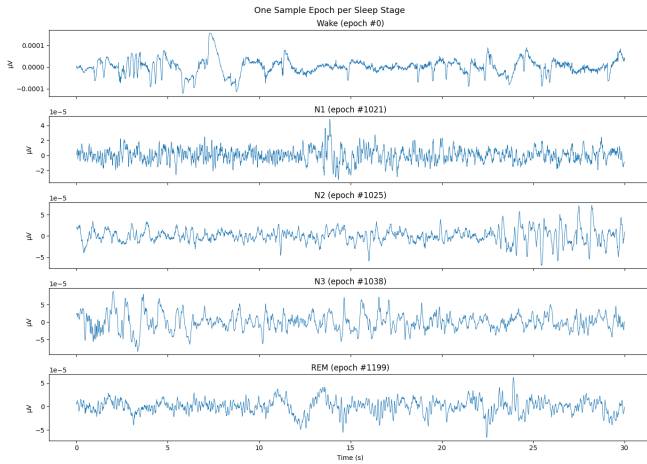


Fig. 1. 30-second EEG epochs for each sleep stage

### III. RESULTS

#### A. Overall Performance

The performance of each of the three models: Support Vector Machine (SVM), Random Forest (RF), and 1D Convolutional Neural Network (1D-CNN), is summarized in Table I. The 1D-CNN achieved the highest performance across all evaluation metrics, with an accuracy of 93.04%, balanced accuracy of 83.3%, and macro F1-score of 79.9%. In comparison, the SVM and RF models achieved lower macro F1-scores

of 72.1% and 74.4%, respectively. These results indicate that the 1D-CNN provides significantly improved performance, particularly when accounting for class imbalance.

TABLE I  
OVERALL MODEL PERFORMANCE COMPARISON

| Model  | Accuracy | Balanced Accuracy | Macro-F1 |
|--------|----------|-------------------|----------|
| SVM    | 86.9%    | 79.6%             | 72.1%    |
| RF     | 92.2%    | 72.0%             | 74.4%    |
| 1D-CNN | 93.0%    | 83.3%             | 79.9%    |

#### B. Per-Class Performance

Per-class F1-scores for each model are shown in Table II. All models performed well on the Wake, N2, and N3 stages, with F1-scores exceeding 80% in most cases. However, performance varied significantly for minority and transitional classes.

TABLE II  
PER-CLASS F1 PERFORMANCE COMPARISON

| Model  | Wake  | N1    | N2    | N3    | REM   |
|--------|-------|-------|-------|-------|-------|
| SVM    | 94.0% | 28.6% | 84.4% | 80.6% | 52.9% |
| RF     | 95.6% | 28.0% | 84.5% | 82.9% | 66.3% |
| 1D-CNN | 98.7% | 37.4% | 88.0% | 86.4% | 77.7% |

The N1 stage was consistently the most difficult to classify, with F1-scores of 28.6% for SVM, 28.0% for RF, and 37.4% for the 1D-CNN. While the 1D-CNN improved N1 performance, it remained the lowest among all classes. In contrast, the REM stage showed substantial improvement with the 1D-CNN, achieving an F1-score of 77.7%, compared to 52.9% and 66.3% for SVM and RF, respectively.

#### C. Confusion Matrix Analysis

The confusion matrix for the 1D-CNN is shown in Fig. 2. It demonstrates that most misclassifications occur between the N1, N2, and REM stages.

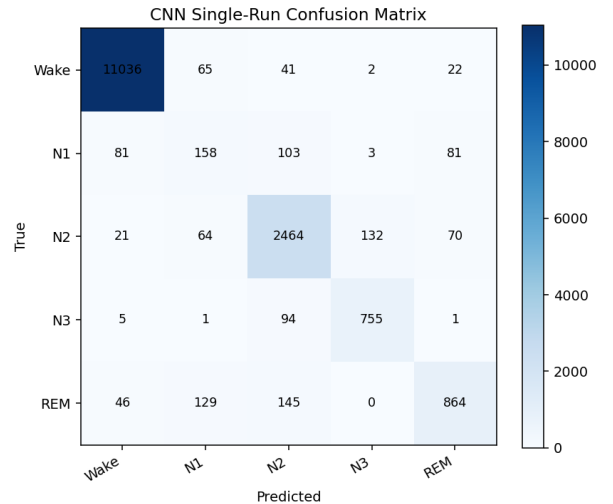


Fig. 2. 1D-CNN Confusion Matrix

This confusion matrix indicates that these classes share very similar EEG characteristics, making them more difficult to distinguish. In contrast, the Wake, N2, and N3 stages were classified with much higher accuracy, with fewer misclassifications across all models.

#### IV. DISCUSSION

The results demonstrate that the 1D-CNN consistently outperforms classical machine learning models across all evaluation metrics, particularly in macro F1-score and balanced accuracy. This indicates improved performance across all sleep stages, including minority classes, rather than gains driven solely by dominant classes such as Wake.

A key challenge observed across all models is the classification of the N1 stage. Despite improvement with the 1D-CNN (F1-score of 37.4%), N1 remains the most difficult class due to its transitional nature between the Wake and N2 stages, as well as its low representation of less than 10% of the dataset. In contrast, the REM stage shows substantial improvement with the 1D-CNN, achieving an F1-score of 77.7%. This evidence suggests that deep learning models are more effective at capturing complex temporal and frequency patterns present in EEG signals. Analyzing the confusion matrix further reveals that most misclassifications occur between N1, N2, and REM stages. These errors are expected, as these stages have overlapping EEG signal characteristics. However, performance on Wake and N3 remains consistently high, proving that these stages are more easily distinguishable.

From a clinical perspective, not all classification errors have equal importance. Accurate identification of N3 (deep sleep) and REM stages is vital, as these stages are associated with physical recovery, memory consolidation, and overall cognitive function. Improvements in these stages, particularly REM, are much more impactful than overall accuracy alone. This highlights the importance of evaluating models using metrics such as macro F1-Score rather than relying solely on accuracy.

Despite the strong performance observed, this project faced several limitations. The dataset exhibits heavy class imbalance, which affects performance on minority classes such as N1. Additionally, only a single EEG channel (Fpz-Cz) was used, which may limit the model's ability to capture more complex physiological patterns. Fpz-Cz in EEG signal analysis is a commonly used channel that measures the voltage difference between the frontopolar midline (Fpz, just above the eyes) and the central midline (Cz, top of the head).

Future work should focus on addressing these limitations by incorporating more complex temporal models such as CNN-LSTM architectures, expanding the overall dataset, and utilizing multiple EEG channels for training. CNN-LSTM hybrid architectures excel at recognizing complex temporal patterns such as sleep stages and often achieve above 90% accuracy in EEG classification. Additional physiological signals could help to improve each model's classification performance as well.

#### V. CONCLUSION

This work presents an automated approach for sleep stage classification using both classical machine learning models and a 1D convolutional neural network. The 1D-CNN achieved the best overall performance, with notable improvements in macro F1-score and balanced accuracy, demonstrating its ability to better handle class imbalances and capture complex EEG signal patterns.

While the model shows strong performance, particularly in the Wake and N2 stages, the N1 stage remains a key challenge to classify due to its transitional and ambiguous nature. These results emphasize that performance evaluation should prioritize class-balanced metrics and clinically meaningful outcomes rather than overall accuracy alone.

From a clinical standpoint, accurate classification of sleep stages such as N3 and REM are essential for assessing sleep quality and supporting diagnosis of sleep disorders. The improvements demonstrated in this work highlight the potential of deep learning methods for scalable and consistent sleep analysis.

Future developments for this project will focus on improving classification of difficult stages such as N1, incorporating temporal modeling techniques, and reconstructing full-night sleep architecture for more comprehensive sleep analysis.

#### Individual Contribution

All data preprocessing, model development, experimentation, evaluation, and report preparation were completed independently by the author, Christian R. Bammann.

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